

End-to-End Azure Data Engineering Project

Overview

This project demonstrates the implementation of a complete End-to-End Azure Data Engineering solution using Azure Data Factory (ADF), Azure Data Lake Storage Gen2 (ADLS Gen2), Azure Databricks, Azure SQL Database, and GitHub. The solution follows the Medallion Architecture (Bronze, Silver, and Gold layers) to ingest, process, transform, and store data in a scalable and analytics-ready format. The project implements incremental data loading to process only newly added or modified records, improving overall performance and reducing processing costs. The final output consists of curated Fact and Dimension tables stored in Parquet format for reporting and analytical workloads.

Architecture

GitHub Repository (Source Data) → Azure Data Factory (ADF) → ADLS Gen2 Bronze Layer → Azure Databricks (PySpark Transformations) → ADLS Gen2 Silver Layer → Azure Databricks (Business Transformations) → ADLS Gen2 Gold Layer → Dimension & Fact Tables (Parquet Format)

Technology Stack

Azure Data Factory (ADF) Azure Data Lake Storage Gen2 (ADLS Gen2) Azure Databricks PySpark Azure SQL Database GitHub Parquet File Format

Data Model

Dimension Tables: DimCustomer, DimProduct, DimDate, DimGeography, DimEmployee Fact Tables: FactSales, FactOrder, FactTransaction

Incremental Loading Strategy

The project uses an incremental loading approach to process only new or modified records instead of reprocessing the entire dataset during every execution. Benefits: - Reduced pipeline execution time - Lower compute costs - Improved performance - Better scalability - Efficient resource utilization

Business Value

This solution demonstrates how modern Azure services can be combined to build scalable, automated, and maintainable data pipelines capable of supporting enterprise-level analytics and reporting requirements.